0 0 0 0
0 0 0 1
0 0 1 0
0 0 1 1
0 1 0 0
0 1 0 1
0 1 1 0
0 1 1 1
1 0 0 0
1 0 0 1
1 0 1 0
1 0 1 1
1 1 0 0
1 1 0 1
1 1 1 0
1 1 1 1

n → 4

$\binom{n}{2} \times 1$

$\binom{n}{4} \times 2$ 2×2

$\frac{10}{4}$

$n = 10$.

[1, 2, 1, 5, 5] [1, 1, 2, 5, 5]

[freq-based Alg. → TC: O(n). AS: O(n).]

— Sorting-based Alg → TC: O(n log n). AS: O(1)

XOR-based Alg → TC: O(n). AS: O(1).

1^2^1^5^5 = 1^1^5^5^2

= 0^0^2

= 0^2 = 2

0 0
XOR 1 0
1 0

XOR
a^a = 0
0^a = a

a^b^c = c^b^a = a^c^b.

ans = 0. [2].

0 1
1 0
1 1

a = 1
b = 3
c = 5.

0 0 1
0 1 1
1 0 1
1 1 1

0 1 1
1 0 1
0 0 1
1 1 1

[1, 10, 1, 1, 3, 7, 7, 7]

→ 0 0 0 1
1 0 1 0 ←
0 0 0 1
0 0 0 1
0 1 1 1
0 1 1 1
0 1 1 1

3x
(3x + 1)

freq: → [1 3 4 6].
1 0 1 0 → 10

[a, a, a, b, b, b, c, c]

a^b^a^a^b.

(4x 3)

[TC = O(n)
AS = O(1).]

num = 1010
= 101
= 10
= 1
= 0

3 2 1 0
1 0 1 0

0 1 2 3
freq: [6, 4, 3, 1]

$(0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$

$\sum_i (\text{freq}[i] \% 3) \times 2^i$

$\sum_i (\text{freq}[i] \% 3)$ ← mod.

4x + 2.
4x + 1

Arrays

ans = 2

0 1 2 3 4
[3, 5, 1, 2, 6]. string

x x+1 x+2 x+3 x+4 x+5

7

ans

Insert
- At end: O(1) → worst case O(n)
At start: O(n).
At middle: O(n).

Delete.
At end: O(1).
At start: O(n).
At middle: O(n).

Random Access: O(1).

[[0, 1, 5, 6]
→ [ ]
5.
[ ]].

1 → 2 → 3 → 8 → 16 → 32 → 64 → 128 → 256 → 512 - - -

[-10, 5, 3, 2, 6].

0/1 = 1

[-10, 2, 3, 5, 1].

[-20, -10, 0, 1, 3]

Sorting-based Alg. : TC: O(n log n) AS: O(1)

Linear Search-based Alg. : TC: O(n²). AS: O(1).

Hash-table-based Alg. : TC: O(n). AS: O(n)

[-10, 0, -20, 1, 3, 2, 5, 7]

1 → 1
3 → 1
2 → 1 → 2
5 → 1